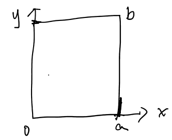
Seperation of variables

§ 13.3

$$\begin{cases} u_{xx} + u_{yy} = 0 \\ u\big|_{x=0} = 0, \quad \frac{\partial u}{\partial x}\Big|_{x=a} = 0 \\ u\big|_{y=0} = f(x) \\ \frac{\partial u}{\partial y}\Big|_{y=b} = 0 \end{cases}$$

$0<x<a \quad 0<y<b$



Seperation of variable approach

$u_{xx} + u_{yy} = 0$

① $u(x,y) = X(x)Y(y)$

$X''(x)Y(y) + X(x)Y''(y) = 0$

$\frac{X''(x)}{X(x)} = -\frac{Y''(y)}{Y(y)} = -\lambda$

$X''(x) + \lambda X(x) = 0 \quad ☆, \quad X(0) = 0 \quad X'(a) = 0$

$Y''(y) - \lambda Y(y) = 0$

② (A) $\lambda = 0$

$X''(x) = 0 \qquad X(x) = Ax + B$

$X(0) = 0 \qquad B = 0, \quad X'(a) = A \cdot a = 0 \quad A = 0$

$X(x) \equiv 0 \qquad u(x,y) \equiv 0 \quad \times$

(B) $\lambda < 0 \qquad \lambda = -s^2 \qquad s = \sqrt{-\lambda}$

$X''(x) - s^2 X(x) = 0$

$X(x) = Ae^{sx} + Be^{-sx}$

$X(0) = A + B = 0 \qquad A = -B$

$X'(a) = A \cdot s e^{sa} - B \cdot s e^{-sa} = 0$

$A e^{sa} - Be^{-sa} = 0$

$-B[e^{sa} + e^{-sa}] = 0$

$A = B = 0 \quad \times$

(C) $\lambda > 0$

$X''(x) + \lambda X(x) = 0 \qquad X(0) = X'(0) = 0$

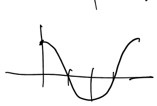
$X(x) = A\cos\sqrt{\lambda}x + B\sin\sqrt{\lambda}x$

$A = 0 \qquad X(0) = 0$

$X'(x) = \qquad + B\sqrt{\lambda}\cos\sqrt{\lambda}x\Big|_{x=a} =$

$B\sqrt{\lambda}\cos\sqrt{\lambda} \cdot a = 0$

$\cos\sqrt{\lambda}a = 0$



$\sqrt{\lambda}a = \frac{\pi(2n+1)}{2}$

eigen value

$\boxed{\sqrt{\lambda} = \frac{2n+1}{2a}\pi} \qquad n = 0, 1, 2, 3 \cdots$

eigen function

$\rightarrow X_n(x) = \sin\frac{2n+1}{2a}\pi x$

③ $Y''(y) - \lambda Y(y) = 0 \qquad \pm\sqrt{\lambda}$

$u\big|_{y=0} = f(x) \quad \frac{\partial u}{\partial y}\Big|_{y=b} = 0$

✓ $Y_n(y) = Ce^{\sqrt{\lambda}y} + De^{-\sqrt{\lambda}y}$

✓ $Y_n(y) = C_n e^{\frac{2n+1}{2a}\pi y} + D_n e^{-\frac{2n+1}{2a}\pi y} \qquad n = 0, 1, 2, 3$

④ $u_n(x,y) = X_n(x)\,Y_n(y)$

$$Y_n(y) = C_n e^{\frac{2n+1}{2a}\pi y} + D_n e^{-\frac{2n+1}{2a}\pi y} \qquad n=0,1,2,3$$

$$u_n(x,y) = X_n(x)\, Y_n(y)$$

$$\begin{cases} u_{xx}+u_{yy}=0 \\ u|_{x=0}=0 \quad \frac{\partial u}{\partial x}\Big|_{x=a}=0 \\ u|_{y=0}=f(x) \quad ? \\ \frac{\partial u}{\partial y}\Big|_{y=b}=0 \quad ? \end{cases}$$

叠加原理
全解

$$\boxed{u(x,y) = \sum_{n=0}^{\infty} \sin\frac{2n+1}{2a}\pi x\left[C_n e^{\frac{2n+1}{2a}\pi y} + D_n e^{-\frac{2n+1}{2a}\pi y}\right]}$$

$$u_y = \sum_{n=0}^{\infty} \sin\frac{2n+1}{2a}\pi x\,\underbrace{\left[C_n \frac{2n+1}{2a}\pi e^{\frac{2n+1}{2a}\pi b} - D_n \frac{2n+1}{2a}\pi e^{-\frac{2n+1}{2a}\pi b}\right]}_{=0} = 0 \qquad ②$$

①

$$u|_{y=0}=f(x):\ \sum_{n=0}^{\infty} \sin\frac{2n+1}{2a}\pi x\,\underbrace{\left[C_n + D_n\right]} = f(x)$$

$$f(x) = \sum_{n=0}^{\infty}\left[C_n+D_n\right]\sin\frac{2n+1}{2a}\pi x$$

$$\boxed{C_n+D_n = \frac{2}{a}\int_0^a f(x)\sin\frac{2n+1}{2a}\pi x\,dx} \quad ①$$

$$\Rightarrow C_n,\ D_n$$

$$\begin{cases} D_n = \frac{2}{a}\int_0^a f(x)\sin\frac{2n+1}{2a}\pi x\,dx \\ C_n = -D_n \tanh\frac{2n+1}{2a}\pi b \end{cases}$$

$$\tanh = \frac{\sinh x}{\cosh x} \qquad \frac{e^x+e^{-x}}{2} = \cosh x$$

$$f(x) = \sum_{n=0}^{\infty} P_n\,\varphi_n(x)$$

Transform Analysis of signals & systems

$$\int_0^{T_0}\varphi_n(x)\varphi_m(x)\,dx = \begin{cases} 0 & n\neq m \\ E & n=m \end{cases}$$

正交 性质 orthogonal, inner product

$$\int_0^{T_0} f(x)\cdot\varphi_2(x)\,dx = \sum_{n=0}^{\infty} P_n\int_0^{T_0}\varphi_n(x)\varphi_2(x)\,dx$$

$$= P_2\cdot E$$

$$P_n = \frac{1}{E}\int_0^{T_0} f(x)\cdot\varphi_n(x)\,dx$$

$$\int_0^a \sin\frac{2n+1}{2a}\pi x\ \sin\frac{2m+1}{2a}\pi x\,dx = \begin{cases} 0 & n\neq m \\ a/2 & n=m \end{cases}$$

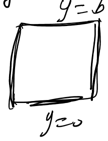
$$= \frac{a}{2}\delta_{nm}$$

$$\delta_{nm} = \begin{cases} 0 & n\neq m \\ 1 & n=m \end{cases}$$

Find C_n D_n

$u(x,y)$

§1.3.4 Heat transfer 热传导



$$\begin{cases} u_t - K(u_{xx}+u_{yy}) = 0 \\ u_x|_{x=0}=0 \quad u_x|_{x=a}=0 \\ u_y|_{y=0}=0 \quad u_y|_{y=a}=0 \\ u|_{t=0}=\phi(x,y) \end{cases}$$

$Z(z)$

$$u(x,y,t) = T(t)\,X(x)\,Y(y)$$

$$u_t - K(u_{xx}+u_{yy}) = 0$$

$$\left(\tfrac{1}{K}\right)(-)\,T'(t)X(x)Y(y) + \left(T(t)X''Y + TXY''\right) = 0$$

$$\frac{X''}{X} + \frac{Y''}{Y} - \frac{1}{K}\frac{T'}{T} = 0$$

$(\frac{1}{K})(-)$ ...

$$\frac{X''}{X} + \frac{Y''}{Y} - \frac{1}{K}\frac{T'}{T} = 0$$

$$\underbrace{\frac{X''}{X} + \frac{Y''}{Y}}_{(x,y)} = \underbrace{\frac{1}{K}\frac{T'}{T}}_{t} = -\lambda$$

$\Rightarrow T' + \lambda K T = 0$

$\Rightarrow X'' + \mu X = 0.$

$\Rightarrow Y'' + \nu Y = 0.$

$$\frac{X''}{X} + \frac{Y''}{Y} = -\lambda$$

$$\frac{X''}{X} = -\lambda - \frac{Y''}{Y} = -\mu$$

$$\frac{Y''}{Y} = \underbrace{+(\mu - \lambda)}_{-\nu} \qquad -\nu = \mu - \lambda$$

$$\underline{\lambda = \mu + \nu}$$

$$\begin{cases} X'' + \mu X = 0 \\ X'(x)\big|_{x=0} = 0 \quad X'(x)\big|_{x=a} = 0 \end{cases} \qquad u_x = \cancel{\nabla}Y\, \underbrace{X'\big|_{x=0}}_{} = 0$$

$$\begin{cases} Y'' + \nu Y = 0 \\ Y'(y)\big|_{y=0} = 0 \quad Y'(y)\big|_{y=a} = 0 \end{cases}$$

② ✓ $\mu = 0$ $\qquad \underline{X(x) = Ax + B} \qquad X(x) = (B)$

$\mu < 0 \qquad X = Ae^{sx} + Be^{-sx}$

$\mu = -s^2 \qquad X' = A \cdot s e^{sx} + B(-s)e^{-sx}$

$s = \sqrt{-\mu}$

$$\begin{cases} A \cdot \; - B \cdot \; = 0 \\ A \cdot \; e^{s \cdot a} - B \; e^{-sa} = 0 \end{cases}$$

$A = B = 0 \qquad \times$

$u > 0$

$$X'' + \mu X = 0$$

$$\underline{X = A\cos\sqrt{u}\,x} + B\sin\sqrt{u}\,x$$

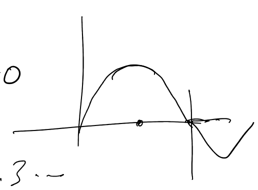
$$X' = -A\sqrt{u}\,\sin\sqrt{u}\,x + B\sqrt{u}\,\cos\sqrt{u}\,x$$

$x = 0 \qquad B\sqrt{u}\,\underbrace{\cos\sqrt{u}\cdot 0}_{1} = 0$

$B = 0$

$$X' = -A\overset{\sqrt{u}}{\,}\sin\sqrt{u}\,x$$

$x = a \qquad X'(a) = -A\overset{\sqrt{u}}{\,}\sin\sqrt{u}\cdot a = 0$



$\sin\sqrt{u}\cdot a = 0$

$\sqrt{u}\cdot a = n\pi \qquad n = 1, 2, 3 \cdots$

$\sqrt{u} = \frac{n\pi}{a} \qquad X(x) = A_n \cos\frac{n\pi}{a}x \quad \leftarrow \underline{n = 1, 2, 3 \cdots}$

$\sqrt{\nu} = \frac{m\pi}{a} \qquad Y(y) = B_m \cos\frac{m\pi}{a}y \quad m = 1, 2, 3 \cdots$

$\underline{T' + \lambda K T = 0} \qquad \lambda = u + \nu$

$e^{-\lambda K t} \qquad = \left(\frac{n\pi}{a}\right)^2 + \left(\frac{m\pi}{a}\right)^2$

$-\lambda K e^{-\lambda K t} + \lambda K e^{-\lambda K t} = 0$

$$T(t) = e^{-\lambda K t} = e^{-K\left[\left(\frac{n\pi}{a}\right)^2 + \left(\frac{m\pi}{a}\right)^2\right]t}$$

$-\lambda k e^{-\lambda k t} + \lambda k e^{-\lambda k t} = 0$

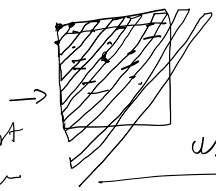
$T(t) = e^{-\lambda k t} = e^{-k[(\frac{n\pi}{a})^2 + (\frac{m\pi}{a})^2]t}$

$u(x,y,t) = \sum T \cdot X \cdot Y \; C_{n,m}$

$\varphi(x,y) = \sum_{n=0}^{\infty}\sum_{m=0}^{\infty} \cos\frac{n\pi}{a}x \, \cos\frac{m\pi}{a}y \cdot e^{-k[(\frac{n\pi}{a})^2 + (\frac{m\pi}{a})^2]t}$

$\varphi(x,y) = u|_{t=0}$

$\varphi(x,y) = \sum_{n=0}^{\infty}\sum_{m=0}^{\infty} C_{n,m}\,\phi_{n,m}(x,y)$ $\quad C_{n,m} = \int_0^a \int_0^b \varphi(x,y)\cos\frac{n\pi}{a}x \cos\frac{m\pi}{a}y \, dx\,dy$



$\sum_{k=0}^{\infty} \cdots$ Zigzag Scan

Hilbert problem Scan

use 1-D strip to fill n-D space

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§13.5

$$\begin{cases} u_{tt} - a^2 u_{xx} = f(x,t) \quad \text{(0)} \quad \text{force} \\ u|_{x=0} = u|_{x=l} = 0 \\ u|_{t=0} = 0,\ g(x) \\ u_t|_{t=0} = 0,\ h(x) \end{cases}$$

受迫振动

$u_{tt} - a^2 u_{xx} = f(x,t)$

$z_{tt} - a^2 z_{xx} = f(x,t)$?

$Au + Bz = V$

$V_{tt} - a^2 V_{xx} = A f(x,t) + B f(x,t)$

$y'' - 6y' + 8y = f(x)$ $\quad$ 0

$y = y_p + y_0$

Find $v(x,t)$ a particular solution

$$\begin{cases} v_{tt} - a^2 v_{xx} = f(x,t) \quad \checkmark \\ v|_{x=0} = 0 \quad v|_{x=l} = 0 \end{cases}$$

let $u(x,t) = v(x,t) + w(x,t)$

$w = u - v$

$$\begin{cases} w_{tt} - a^2 w_{xx} = 0 \\ w|_{x=0} = w|_{x=l} = 0 \quad \checkmark \\ w|_{t=0} = u|_{t=0} - v|_{t=0} = -v(x,0) = \varphi(x) + g(x) \\ w_t|_{t=0} = u_t|_{t=0} - v_t|_{t=0} = -v_t(x,0) = \psi(x) + h(x) \end{cases}$$

Example $\quad f(x,t)$

Example $\qquad f(x,t)$

$$u_{tt} - a^2 u_{xx} = A_0 \sin\omega t$$

$$u|_{x=0} = u|_{x=l} = 0$$

$$u|_{t=0} = u_t|_{t=0} = 0$$

$$\begin{cases} V_{tt} - a^2 V_{xx} = A_0 \sin\omega t. \\ V|_{x=0} = V|_{x=l} = 0 \end{cases}$$

assume $V(x,t) = f(x)\sin\omega t$ $\qquad \sin\omega t$

$$f(x)(-\omega^2)\sin\omega t - a^2 f''(x) = A_0 \sin\omega t$$

$$-\omega^2 f(x) - a^2 f''(x) = A_0$$

$$f''(x) + \frac{\omega^2}{a^2} f(x) = -\frac{A_0}{a^2}$$

$$f''(x) + \frac{\omega^2}{a^2} f(x) = 0$$

ω $\boxed{\begin{matrix} f(0)=0 \\ f(l)=0 \end{matrix}}$ $\Downarrow$ $\qquad A=0$

$B \sin\frac{\omega}{a} l = 0$

$$f_p(x) = C \qquad \frac{\omega^2}{a^2} C = -\frac{A_0}{a^2}$$

$$C = -A_0/\omega^2$$

$$f(x) = A\cos\frac{\omega}{a}x + B\sin\frac{\omega}{a}x + \frac{-A_0}{\omega^2}$$

$$A.B \begin{cases} A = \frac{A_0}{\omega}\tan\frac{\omega l}{2a} \\ B = \frac{A_0}{\omega^2} \end{cases}$$

$$V(x,t) = f(x)\sin\omega t \quad / \ \omega f(x)\cos\omega t$$

$$u(x,t) = V(x,t) + w(x,t)$$

$$w = u - v$$

$$\begin{cases} w_{tt} - a^2 w_{xx} = 0 \\ w|_{x=0} = w|_{x=l} = 0 \\ w|_{t=0} = -V(x,0) = 0 \quad = 0 \\ w_t|_{t=0} = -V_t(x,0) = -\omega f(x) = \psi(x) \end{cases}$$

$$w(x,t) = \sum_{n=1}^{\infty}\left(C_n \sin\frac{n\pi}{l}at + D_n\cos\frac{n\pi}{l}at\right)\sin\frac{n\pi}{l}x$$

$$C_n = \frac{2}{n\pi a}\int_0^l \psi(x)\sin\frac{n\pi}{l}x\,dx$$

$$D_n = \frac{2}{l}\int_0^l \phi(x)\sin\frac{n\pi}{l}x\,dx = 0$$

$$C_n = \frac{2}{n\pi a}\int_0^l -\omega\left[\cos\frac{\omega}{a}x \cdot A + B\sin\frac{\omega}{a}x + \frac{-A_0}{\omega^2}\right]\sin\frac{n\pi}{l}x\,dx$$

$$C_n = -\frac{2A_0\omega l^3}{\pi^2 a}\,\frac{1-(-1)^n}{n^2}\,\frac{1}{(n\pi a)^2 - (\omega l)^2}$$

$$\begin{cases} u_{tt} - a^2 u_{xx} = 0 \\ u|_{x=0} = \underline{\mu(t)} \qquad u|_{x=l} = \nu(t) \\ u|_{t=0} = 0 \qquad u_t|_{t=0} = 0 \end{cases}$$

$$u(x,t) = v(x,t) + w(x,t)$$

齐次化
函数

$$\begin{cases} v_{tt} - a^2 v_{xx} = 0 \\ v|_{x=0} = \mu(t) \qquad v|_{x=l} = \nu(t) \end{cases} \qquad w = u - v$$

$$\begin{cases} w_{tt} - a^2 w_{xx} = 0 \\ w|_{x=0} = w|_{x=l} = 0 \\ w|_{t=0} = -v(x,0) \qquad w_t|_{t=0} = -v_t(x,0) \end{cases}$$

$$v(x,t) = \frac{1}{l}[\nu(t) - \mu(t)]x + \underline{\mu(t)}$$

Ex 13.3 $\quad u_{tt} - k u_{xx} = 0$

$$\begin{cases} u|_{x=0} = A\sin\omega t \qquad u|_{x=l} = 0 \\ u|_{t=0} = 0 \qquad u_t|_{t=0} = 0 \end{cases}$$

$$v(x,t) = A\left(1 - \frac{x}{l}\right)\sin\omega t$$

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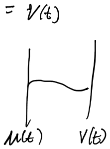
§13.6

$$\begin{cases} u_{tt} - a^2 u_{xx} = 0 \\ u|_{x=0} = \mu(t) \qquad u|_{x=l} = \nu(t) \\ u|_{t=0} = 0 \qquad u_t|_{t=0} = 0 \end{cases}$$

$$S(x,t)|_{x=0} = \mu(t) \qquad S(x,t)|_{x=l} = \nu(t)$$

$$S(x,t) = \mu(t) + \frac{1}{l}[\nu(t) - \mu(t)]\cdot x$$



$$w(x,t) = \underline{u(x,t)} - \underline{S(x,t)}$$

$$w_{tt} - a^2 w_{xx} = \underbrace{(u_{tt} - a^2 u_{xx})}_{0} - \underline{(S_{tt} - a^2 S_{xx})} = f(x,t)$$

$$\begin{cases} w_{tt} - a^2 w_{xx} = f(x,t) \\ w|_{x=0} = 0 \qquad w|_{x=l} = 0 \end{cases}$$

Ex 13.3

$u_{tt} - k u_{xx} = 0$

Ex 13.3

$$\begin{cases} u_t - k u_{xx} = 0 \\ u|_{x=0} = A\sin\omega t \qquad u|_{x=l} = 0 \\ u|_{t=0} = 0 \end{cases}$$

$$S(x,t) = A(1-\tfrac{x}{l})\sin\omega t$$

$$w(x,t) = u(x,t) - S(x,t)$$

$$w_t - k w_{xx} = (u_t - k u_{xx}) - (S_t - kS_{xx})$$

$$= 0 - (\omega A(1-\tfrac{x}{l})\cos\omega t - k\cdot 0)$$

$$\begin{cases} w_t - k w_{xx} = -A\omega(1-\tfrac{x}{l})\cos\omega t \\ w|_{x=0} = 0 \qquad w|_{x=l} = 0 \\ w|_{t=0} = 0 \end{cases} \quad f(x,t)$$

$$w(x,t) = X(x)\,T(t)$$

Assume

$$\begin{cases} w(x,t) = \sum_{n=1}^{\infty} T_n(t)\, X_n(x) \\ f(x,t) = \sum_{n=1}^{\infty} g_n(t)\, X_n(x) \end{cases}$$

$$w_t - k w_{xx}$$

$$= \sum_{n=1}^{\infty} T_n'(t)\, X_n(x) - k\sum_{n=1}^{\infty} T_n(t)\, X_n''(x) = \sum_{n=1}^{\infty} g_n(t)\, X_n(x)$$

Assume $\begin{cases} X_n''(x) + \lambda_n X_n(x) = 0 \\ X_n(0) = X_n(l) = 0 \end{cases} \Rightarrow X_n(x) = \sin\frac{n\pi}{l}x$

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2$$

$$X_n''(x) = -\lambda_n X_n(x)$$

$$\sum_{n=1}^{\infty} T_n'(t)\, X_n(x) - k\sum_{n=1}^{\infty} T_n(t)(-\lambda_n)X_n(x)$$

$$= \sum_{n=1}^{\infty} g_n(t)\, X_n(x)$$

$$\sum_{n=1}^{\infty}\left[T_n'(t) + \lambda_n k\, T_n(t)\right] X_n(x) = \sum_{n=1}^{\infty} g_n(t)\, X_n(x)$$

$$\begin{cases} T_n'(t) + \lambda_n k\, T_n(t) = g_n(t) \\ y' + Py = Q \\ T_n(t)|_{t=0} = 0 \end{cases}$$

$$f(x,t) = A\omega(1-\tfrac{x}{l})\sin\omega t \qquad 1-\frac{x}{l} = \sum_{n=1}^{\infty} C_n \sin\frac{n\pi}{l}x$$

$$f(x,t) = \sum_{n=1}^{\infty} g_n(t)\sin\frac{n\pi}{l}x$$

$$f(x,t) = \left(\sum_{n=1}^{\infty} C_n \sin\frac{n\pi}{l}x\right) \sin\omega t \cdot A\omega$$

$$\boxed{1-\frac{x}{l} = \sum_{n=1}^{\infty} \frac{2}{n\pi} \sin\frac{n\pi}{l}x}$$

$$f(x,t) = \sum_{n=1}^{\infty} \sin\frac{n\pi}{l}x \,\Big|\, \sin\omega t \cdot A\omega \cdot \frac{2}{n\pi}$$

$$T_n''(t) + \left(\frac{n\pi}{l}\right)^2 K\, T_n(t) = \frac{2}{n\pi} A\omega \sin\omega t$$

$$y' + Py = Q \qquad I = \int p\,dx$$

$$y =$$

(13.86)

$$T_n(t) =$$

Ex 13.4

$$\begin{cases} u_{tt} - a^2 u_{xx} = 0 \\ u|_{x=0} = 0 \quad u_x|_{x=l} = A\sin\omega t \\ u|_{t=0} = 0 \quad u_t|_{t=0} = 0 \end{cases}$$

构建 $v(x,t) = f(x)\sin\omega t$ construct

$$v_{tt} - a^2 v_{xx} = f(x)(-\omega^2)\sin\omega t - a^2 f''(x)\sin\omega t = 0$$

$\omega f(x) \cos\omega t$

$$\begin{cases} a^2 f''(x) + \omega^2 f(x) = 0 \\ f''(x) + \left(\frac{\omega}{a}\right)^2 f(x) = 0 \\ f(0) = 0 \quad f(l) = A \end{cases}$$

$$\Downarrow$$

$$f(x) = \frac{Aa}{\omega} \frac{1}{\cos(\omega l/a)} \sin\frac{\omega}{a}x$$

$$\Rightarrow \begin{cases} V_{tt} - a^2 V_{xx} = 0 \\ V|_{x=0} = 0 \quad V|_{x=l} = A\sin\omega t \\ V|_{t=0} = 0 \quad V_t|_{t=0} = \omega f(x) \end{cases}$$

let $w = u - v$

$$\begin{cases} w_{tt} - a^2 w_{xx} = 0 \\ w|_{x=0} = 0 \quad w_x|_{x=l} = 0 \end{cases}$$

$$\begin{cases} W_{tt} - a^2 W_{xx} = 0 \\ W|_{x=0} = 0 \quad W_x|_{x=l} = 0 \\ W|_{t=0} \quad W_t|_{t=0} = -\omega f(x) \end{cases}$$

$$W(x,t) = \sum_{n=0}^{\infty} \underline{\left(C_n \sin \frac{2n+1}{2l} \pi a t + D_n \cos \frac{2n+1}{2l} \pi a t\right)} \underline{\sin \frac{2n+1}{2l} \pi x}$$

$$W = T \cdot X$$

$\lambda = 0$

$\lambda < 0 \quad \lambda > 0$

$$\begin{cases} C_n = \quad (13.92) \\ D_n = 0 \end{cases}$$

常数变易法

variation of constants

$y' + py = Q \qquad I = \int p\,dx$

$y = e^{-I} \int Q e^{I} dx + c e^{-I}$

$y'' + py' + qy = \underline{f(x)}$

$y''(x) + p\, y'(x) + q\, y(x) = f(x)$

p, q are constants

$y'' + py + q = 0 \qquad \lambda_1\ \lambda_2$

$\lambda^2 + p\lambda + q = 0 \qquad \lambda_1\ \lambda_2$

$y(x) = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$

$y(x) = c_1\, y_1(x) + c_2\, y_2(x)$

$y'' + py' + qy \overset{0}{\neq} f(x)$ ✓

$y(x) = c_1(x)\, y_1(x) + c_2(x)\, y_2(x)$

$y' = c_1' y_1 + c_1 y_1' + c_2' y_2 + c_2 y_2'$

$= \underbrace{(c_1' y_1 + c_2' y_2)}_{0} + (c_1 y_1' + c_2 y_2')$

Assume $c_1' y_1 + c_2' y_2 = 0$

$y' = c_1 y_1' + c_2 y_2'$

$y'' = c_1' y_1' + c_1 y_1'' + c_2' y_2' + c_2 y_2''$

$y'' + py' + qy$

$= c_1' y_1' + c_1 y_1'' + c_2' y_2' + c_2 y_2''$

$+ p(c_1 y_1' + c_2 y_2') + q(c_1 y_1 + c_2 y_2)$

$= c_1 y_1'' + c_1 p y_1' + c_1 q y_1 \quad \to 0$

$+ c_2 y_2'' + c_2 p y_2' + c_2 q y_2 \quad \to 0$

$+p(c_1 y_1' + c_2 y_2') + q(c_1 y_1 + c_2 y_2)$

$= c_1 y_1'' + c_1 p y_1' + c_1 q y_1 \to 0$

$+ c_2 y_2'' + c_2 p y_2' + c_2 q y_2 \to 0$

$+ c_1' y_1' + c_2' y_2' = f(x)$

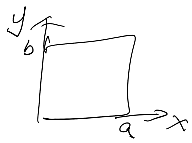
$$\begin{cases} c_1' y_1 + c_2' y_2 = 0 \\ c_1' y_1' + c_2' y_2' = f(x) \end{cases} \qquad \begin{array}{l} y_1 = e^{\lambda_1 x} \\ y_2 = e^{\lambda_2 x} \end{array}$$

$\Rightarrow \quad c_1' = \underline{\qquad} \qquad c_1 = \int \underline{\quad}\, dx$

$\quad\;\; c_2' = \underline{\qquad} \qquad c_2 = \int \underline{\quad}\, dx$

$u = X(x) \cdot T(t) \qquad X(x)\, Y(y)$

$f(x,t) \neq 0$



$$\textcircled{0}\begin{cases} u_{xx} + u_{yy} = f(x,y) \\ u|_{x=0} = \xi(y) \qquad u|_{x=a} = \eta(y) \\ u|_{y=0} = \phi(x) \qquad u|_{y=b} = \psi(x) \end{cases}$$

$$\overset{u}{V}(x,y) = \eta(y) + \underbrace{\left(1 - \frac{x}{a}\right)}_{f(x)} \left[\xi(y) - \eta(y)\right]$$

$f(0) = 1 \qquad f(a) = 0$

$\underline{W = u - V}$